

Problem 1

A Map of the Day-Night Contrast of the Extrasolar Planet HD189733b

- a. Hot Jupiters are assumed to be tidally locked, the same face shows during the whole orbit, making the measure of their temperature distribution an interesting study given permanent day and night sides. The team conducted a 33 hour observing campaign monitoring HD189733b, spanning slightly more than half of the planet's orbit. With quality data, they estimate a brightness temperature of 1200 K on the dayside with a brightness temperature of 970 on the nightside. Their observations imply that the time scale for altering temperature by radiation (solar light re-emitted) is greater than the time scale for homogenizing the temperature, so the cold side isn't much colder than the dayside.

The Broadband Infrared Emission Spectrum of the Exoplanet HD 189733b

- b. Measurements of HD189733b as it passes behind the host star. At infrared wavelengths, the planet to star ratio improves such that the emitted radiation can be studied (or brightness) and such was done with Spitzer observations across various bands. Predicted in theory, they find that their data varies from the Planck curve, and the spectrum is actually twice as great as expected from a blackbody. They attribute this to absorption from both water and carbon monoxide in adjacent wavelengths. This points them to a theory on a distinction between hot Jupiters with or without temperature inversion, cooler air at lower levels.

The Deep Blue Color of HD189733b: Albedo Measurements with HST/STIS at Visible Wavelengths

- c. Similar to measurements of the emission spectrum, they observe HD189733b as it passes behind the star. This time, in visible wavelengths where the brightness still changes like that in the infrared, to derive the planet's geometric albedo (a measure of brightness). If clouds are present, the blue channel can be relatively high, and that is what their models suggest, that it would be visible in blue, in comparison to the red channel.

Retrievals on NIRCam Transmission and Emission Spectra of HD 189733b with PLATON 6, a GPU Code for the JWST Era

- d. First, a retrieval code is a code that instead of producing something like spectra from parameters, the parameters are inferred from observations. They use PLATON to retrieve parameters of HD189733b like metallicity, % ratio, etc. The observations are observations from the near infrared camera of the JWST.